

1.8 Classwork: Compound interest and inflation — Markscheme

1. [Maximum mark: 6]

Sofia invests \$5000 at 4% per year compounded annually.

(a) 1.04 A1

Accept $1 + \frac{4}{100}$. A multiplier of 0.04 or 1.4 earns no mark.

(b) 5000×1.04^6 M1

$= \$6326.595 \dots \approx \6326.60 A1

Accept \$6330 (3 s.f.), \$6327 or \$6326.60. Accept the TVM solver: $N = 6$, $I\% = 4$, $PV = -5000$, $PMT = 0$, $P/Y = C/Y = 1$.

An exponent of 5 (giving \$6083.26) earns the M1 only. Simple-interest working ($5000 + 6 \times 200 = 6200$) earns no marks.

(c) $6326.60 - 5000 = \$1326.60$ A1

Accept \$1330 (3 s.f.) or \$1327.

FT: award A1 for the candidate's own value from (b) minus \$5000.

(d) $5000 \times 1.04^n > 8000$ M1

After 11 years: \$7697.27 (< 8000); after 12 years: \$8005.16 (> 8000)

12 years A1

Accept $1.04^n > 1.6$ solved on the GDC or with logarithms: $n > 11.98 \dots$, so 12 years. Accept the TVM solver for N with $PV = -5000$, $FV = 8000$.

Accept a table search: full marks for showing both adjacent years and stating 12. A single row (after 12 years the value is \$8005.16, so 12) still earns full marks, with the feedback note that the adjacent year 11 is what justifies "first". See markscheme-spec §18.

11.98 is not an acceptable final answer — the answer must be a whole number of years. Rounding down to 11 earns the M1 only.

2. [Maximum mark: 7]

\$12 000 invested at 5.2% per year compounded monthly for 10 years. Inflation 2.8% per year.

$$(a) \quad 12\,000 \left(1 + \frac{5.2}{100 \times 12}\right)^{12 \times 10} \quad \text{M1}$$

$$= \$20\,161.668 \dots \approx \$20\,161.67 \quad \text{A1}$$

Accept \$20 200 (3 s.f.), \$20 162 or \$20 161.67. Accept the TVM solver: $N = 120$, $I\% = 5.2$, $PV = -12\,000$, $PMT = 0$, $P/Y = C/Y = 12$.

M1 for a monthly rate of $\frac{5.2}{1200}$ (or 0.004333...) and an exponent of 120. Using 5.2% compounded annually (giving \$19 922.26) earns no marks.

$$(b) \quad \frac{20\,161.67}{1.028^{10}} \quad \text{M1}$$

$$= \$15\,296.61 \dots \approx \$15\,296.61 \quad \text{A1}$$

Accept \$15 300 (3 s.f.) or \$15 297.

Accept the real-interest-rate method: $5.2 - 2.8 = 2.4\%$ per year compounded monthly, $12\,000 \left(1 + \frac{2.4}{1200}\right)^{120} = \$15\,251.33$, which also rounds to \$15 300 (3 s.f.). Accept the TVM solver with $I\% = 2.4$.

Multiplying by 1.028^{10} (giving \$26 574) earns no marks: the real value must be less than the value in (a). Subtracting 2.8% once ($0.972 \times 20\,161.67 = 19\,597$) earns no marks.

FT: award M1A1 for the candidate's own value from (a) divided by 1.028^{10} .

$$(c) \quad 15\,296.61 - 12\,000 = \$3\,296.61 \quad \text{A1}$$

Accept \$3300 (3 s.f.) or \$3297. With the real-rate method, \$3251.33 (\$3250 to 3 s.f.) is accepted.

$20\,161.67 - 12\,000 = 8161.67$ earns no mark: that is the increase before inflation is taken into account.

FT: award A1 for the candidate's own value from (b) minus \$12 000.

$$(d) \quad \frac{12\,000}{1.028^{10}} \quad \text{M1}$$

$$= \$9104.374 \dots \approx \$9104.37 \quad \text{A1}$$

Accept \$9100 (3 s.f.) or \$9104. Accept the real-rate method: $12\,000 \times 0.972^{10} = \9033.25 (\$9030 to 3 s.f.).

The cash keeps its face value of \$12 000 but loses about a quarter of its buying power. "\$12 000" earns no marks.